

Equation of State for Solid Phase I of Carbon Dioxide Valid for Temperatures up to 800 K and Pressures up to 12 GPa

J. P. Martin Trusler^{a)}

Department of Chemical Engineering, Imperial College London, South Kensington Campus, London SW7 2AZ, United Kingdom

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The available thermodynamic-property data for solid phase I of carbon dioxide (“dry ice”) are reviewed and used to determine the parameters of a new fundamental equation of state constructed in the form of a Helmholtz energy function with temperature and molar volume as the independent variables. The experimental data considered include the pressure, molar volume, and isobaric heat capacity along the sublimation curve, the melting-pressure curve, and molar volume in the compressed solid at temperatures from 295 to 764 K and pressures up to 12 GPa. The equation of state is based on the quasi-harmonic approximation, incorporating a Debye oscillator distribution for the vibrons, two discrete modes for the librons and a further three distinct modes for the internal vibrations of the CO₂ molecule. A small anharmonic correction term is included, which is significant mainly in the region of the triple point. The estimated relative uncertainty of molar volume at specified temperature and pressure calculated from the equation of state is 0.02% on the sublimation curve and 1.5% in the compressed solid; for isobaric heat capacity on the sublimation curve, the uncertainty varies from 5.0% to 0.5% between 2 and 195 K. Auxiliary equations for the pressure and molar volume on the sublimation and melting curves are given. The equation of state is valid at temperatures from 0 to 800 K and at pressures from the solid–fluid phase boundary to 12 GPa. © 2011 American Institute of Physics. [doi:10.1063/1.3664915]

Key words: CO₂; compressibility; density; equation of state; expansivity; heat capacity; Helmholtz energy; melting curve; molar volume; solid carbon dioxide; sublimation curve.

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^{a)}Electronic mail: m.trusler@imperial.ac.uk..

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1. Introduction

Carbon dioxide is a very common and well-studied substance. On earth, it is found in the atmosphere, oceans, upper mantle, and crust, and has profound effects in each. CO_2 is also found in the atmosphere or, as solid, on the surfaces of some planets. Interests in the solid phases of CO_2 in the fields of earth and planetary sciences divide between the high-temperature high-pressure properties, relevant in the upper mantle,¹ and low-temperature behavior, relevant to surface deposits of CO_2 on cold planets and their satellites.^{2,3} Carbon dioxide is also of major technological importance and the demands of large-scale CO_2 capture, transport, and storage call for an understanding of all the phases of this substance.⁴ In particular, the solid is likely to form in the event of rupture of or leakage from high-pressure transmission lines operating with CO_2 , and knowledge of the thermodynamic properties is required to model such events and to develop safety cases. The normal solid (“dry ice”) is scientifically interesting as the classical example of an orientationally ordered molecular crystal, and there is also much interest in the other phases of solid CO_2 , which are observed under extreme compression.

As a molecule, CO_2 is exceptionally stable and characterized by a rigid linear structure having a substantial electric quadrupole moment. The intermolecular forces between pairs of CO_2 molecules are dictated largely by short-range repulsions, dispersion forces, and the quadrupole–quadrupole interaction, all of which are anisotropic in nature. The quadrupole moment is thought to be responsible for the relatively high triple-point and critical pressures of CO_2 (0.517 MPa and 7.38 MPa, respectively), and the interplay of the different intermolecular forces gives rise to polymorphism in the solid phase. The thermodynamic properties of the fluid phases have been extensively studied and are represented at temperatures up to 1100 K and pressures up to 800 MPa by a reference-quality equation of state.⁵ Many properties of dry ice have been measured, but there exists no comprehensive thermodynamic model that reconciles all of the available data. This can be regarded as a gap in present scientific knowledge of the nature of solid CO_2 I and also as a handicap in engineering calculations related to processes in which solid CO_2 is formed intentionally or accidentally. Thus the main purpose of the present work was to provide a soundly based thermodynamic model from which the properties of solid CO_2 I may be obtained over the full temperature and pressure ranges of its existence.

2. Phase Diagram

Determination of the thermodynamic phase diagram of CO_2 is complicated by metastability and kinetic barriers to solid–solid phase transitions, but a reasonably clear picture of the gross features has emerged and is depicted in Fig. 1.^{6,7} At temperatures below about 800 K, application of pressure causes the fluid to crystallize into the low-pressure dry-ice form of CO_2 , designated as phase I, which is a simple

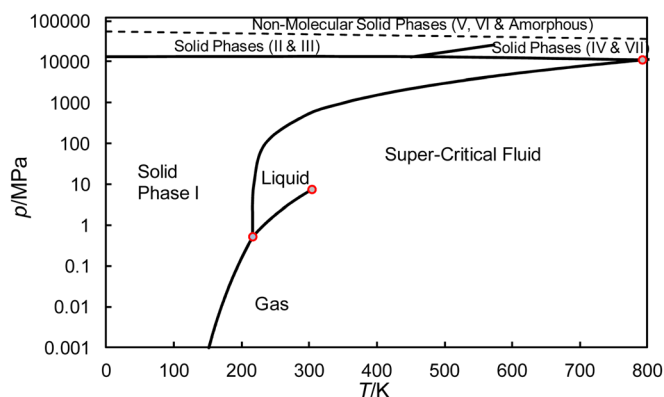


FIG. 1. (Color online) Phase diagram of CO₂ from Refs. 5–7. Circles highlight triple points and the gas–liquid critical point.

orientation-ordered soft molecular crystal with the cubic Pa3 structure.⁶ This phase is stable for $T \leq 800$ K at pressures up to around 10 to 13 GPa, above which it transforms into other crystalline structures, designated II, III, IV, and VII, in which the CO₂ molecule persists as an entity. Phase III is most probably metastable with respect to phases II and IV, but the precise details of the phase boundaries between phases I, II, III, IV, and VII are subject to considerable uncertainty. From the melting-curve data,^{8,9} it is inferred that a triple point exists at $T \approx 800$ K and $p \approx 11$ GPa at which fluid and solid phases I coexist with either phase IV or VII. At higher pressures, the molecular-crystalline phases undergo transitions to non-molecular forms in which CO₂ units are no longer the building block. These phases include the quartz-like phase V and amorphous solids. Eventually, dissociation into diamond and molecular oxygen is thought to occur at $T > 1500$ K and $p > 30$ to 80 GPa.^{1,6}

3. Previous Thermodynamic Models

No fundamental equation fitted to a wide range of experimental data has yet been published. A number of modeling studies using intermolecular potentials of increasing complexity have been reported^{10–16} involving consideration of thermodynamic properties, lattice vibrations, and phase transitions of solid CO₂. It is quite possible that an intermolecular potential model could be sufficiently refined to describe all of these properties to within experimental error. However, such an approach would not be easy to apply, as stochastic techniques are required to obtain the observable properties of the system. Giordano *et al.*¹⁷ derived a closed-form thermal equation of state on the basis of experimental pressure–volume–temperature data. Their equation provides an excellent description of the high-pressure compression data at $T \geq 295$ K and a good description of the molar volume along the sublimation curve. However, caloric properties were not considered and so, as it stands, the equation of state cannot be used to calculate heat capacity or other related properties. Span and Wagner⁵ developed a correlation for the sublimation pressure that was fitted to critically assessed data at $T \geq 154$ K and is characterized by uncertainties of ≤ 250 Pa. A correlation of the melting curve

for $T \leq 266$ K was also presented. In 2006, Giordano *et al.*⁸ reported a correlation of the melting curve, based on new measurements, valid for temperatures up to 800 K.

The objective of the present work was to represent all of the available experimental data, ideally to within the experimental uncertainty, by means of a fundamental equation of state from which all thermodynamic properties of solid phase I could be computed. When coupled with an equation of state for the fluid phases, such a model would also permit calculation of the solid–fluid phase boundaries.

4. Available Experimental Data

Many thermodynamic properties of solid phase I of carbon dioxide have been studied experimentally to some extent. However, the amount of available data is limited, and the distribution of states investigated is not as wide as might be desired. Accordingly, it is necessary to consider all the available data carefully and to include even some rather old work in order to have sufficient information to derive a wide-ranging equation of state. Ideally, experiments would be performed on single crystals, free of defects, but many of the reported data relate to polycrystalline samples and this may be responsible for some of the discrepancies that exist between some data sources.

The collected data have been assigned to one of three groups. Group 1 contains data tabulated in peer-reviewed sources that either include an assessment of the experimental uncertainties or provide sufficient information to support reasonable estimates thereof. Group 1 data were used in the optimization of the parameters in the equation of state. Data in group 2 were read from graphs; or lack a reliable estimate of uncertainty; or pertain to the same property and range of states as a group 1 dataset having lower uncertainty; or require for their interpretation other thermodynamic properties that may not be known with commensurate accuracy. Nevertheless, data from group 2 are useful as a means of verification. A few of the datasets found were thought to be unreliable; these were assigned to group 3 and not considered further.

4.1. Data on the sublimation curve

The properties of solid carbon dioxide have been well studied along the sublimation curve from temperatures as low as 2 K up to the triple point. The available single-phase data are summarized in Table 1^{18–30} and the phase-change data are summarized in Table 2.^{24,31–38} The most accurate measurements of the molar volume V_m are the x-ray diffraction data of Krupskii *et al.*,²⁹ which extend from 6 K up to 200 K and have a relative uncertainty of 0.01%. These authors also derived the isobaric expansivity α by differentiation and provided extrapolated values of V_m and α extending up to the triple-point temperature. The extrapolation was based on the principle of corresponding states with solid argon as the reference substance. Much earlier, Keesom and Kohler^{22,23} used x-ray diffraction to determine V_m along the sublimation curve. Their original results have been corrected

TABLE 1. Summary of single-phase data on the sublimation curve

Property	Source	Year	T/K	u_r^a	Group
V_m	Maass and Barnes ¹⁹	1926	90 to 192	0.2%	2
V_m	Heuse ²¹	1930	20	0.2%	2
V_m	Keesom and Kohler ^{22,23,27}	1934	20 to 114	0.1% ^b	2
V_m	Krupskii <i>et al.</i> ²⁹	1982	6 to 216	0.01%	1
α	Krupskii <i>et al.</i> ²⁹	1982	6 to 216	5% to 10%	2
α	Manzhelii <i>et al.</i> ²⁶	1971	13 to 139	2% to 10%	1
$C_{p,m}$	Eucken ¹⁸	1916	20 to 45		2
$C_{p,m}$	Maass and Barnes ¹⁹	1926	109 to 183	0.2%	2
$C_{p,m}$	Eucken and Hauck ²⁰	1928	80 to 210		2
$C_{p,m}$	Giauque and Egan ²⁴	1937	15 to 195	0.2% to 3%	1
$C_{p,m}$	Manzhelii <i>et al.</i> ²⁶	1971	2.1 to 38	1% to 5%	1
κ_S	Hovi and Mäntylä ²⁵	1959	195		3
κ_S	Manzhelii <i>et al.</i> ²⁶	1971	88 to 190	4%	2
κ_S	Powell <i>et al.</i> ²⁸	1972	95	8%	2
κ_S	Yamashita and Kato ^{30c}	1997	77 to 195	6% ^b	2

^a u_r denotes relative standard uncertainty.^bEstimated.^cData reported graphically only.

for a small error in the x-ray wavelength as detailed by Curzon.²⁷ The isobaric molar heat capacity $C_{p,m}$ has been measured by Giauque and Egan²⁴ from 15 K to the normal sublimation temperature, and by Manzhelii *et al.*²⁶ from 2 to 39 K; both sources were included in group 1. Other published data for $C_{p,m}$ ^{18–20} were assigned to group 2. Manzhelii *et al.*²⁶ measured the isobaric expansivity from 13 to 139 K by pycnometry and derived the isentropic compressibility κ_S at temperatures between (88 and 190) K from measurements of the longitudinal and transverse speeds of sound in a polycrystalline sample of solid CO₂. The 1% relative uncertainty of the speeds of sound claimed by these authors corresponds to a relative uncertainty of 4% in κ_S . Manzhelii *et al.* extrapolated their sound speeds down to zero temperature, assuming a quadratic dependence upon T , and thereby obtained estimates of κ_S in this low-temperature region also. The Debye temperature obtained from the extrapolated longitudinal and transverse sound speeds in the limit $T \rightarrow 0$ K, $\theta_D = 153$ K, agrees with that obtained from their low-

temperature heat-capacity data. Yamashita and Kato also report values of the ultrasonic speeds in a similar temperature range.³⁰ Unfortunately, the data are presented only graphically and, from the scatter, the derived values of κ_S must have a relative uncertainty of about 6%. Furthermore, the results are systematically lower than those of Manzhelii *et al.*²⁶ by between 8 and 13% between 90 and 190 K. Yamashita and Kato argue that their lower values of κ_S must be more reliable, as the presence of cracks, voids, and other imperfections in polycrystalline samples serve to increase the compressibility. However, Yamashita and Kato's measurements for solid methane also imply smaller values of κ_S than reported in other literature.³⁹ As the discrepancies between these two data sources cannot be resolved from the published details, both were assigned to group 2. Two other sources are known from which κ_S may be determined: the elastic constants of a single crystal of solid CO₂ were obtained at $T = 95$ K by Powell *et al.*²⁸ from inelastic neutron scattering data with an uncertainty of 8%; and the

TABLE 2. Summary of phase-change data on the sublimation curve

Property	Source	Year	T/K	$u(T)^a/K$	$u(p)^a/\text{Pa}$	u_r^a	Group
p_{sub}	Meyers and Van Dusen ³²	1933	194 to 216			0.02%	2
p_{sub}	Giauque and Egan ²⁴	1937	195	0.02	3		2
p_{sub}	Tickner and Lossing ³³	1951	106 to 154	0.3			2
p_{sub}	Ambrose ³⁴	1956	178 to 198	0.003	40		1
p_{sub}	Baughman <i>et al.</i> ³⁵	1974	202 to 212	0.01	200		2
p_{sub}	Bryson <i>et al.</i> ³⁶	1974	67 to 111			3%	2
p_{sub}	Bilkadi <i>et al.</i> ³⁷	1975	154 to 216	0.02	10		1
p_{sub}	Fernandez-Fassnecht and del Rio ³⁸	1984	194 to 216	0.001	100		1
$\Delta_s^s H_m$	Maass and Barnes ¹⁹	1926	195				2
$\Delta_s^s H_m$	Eucken and Donath ³¹	1926	140 to 170			0.2%	2
$\Delta_s^s H_m$	Giauque and Egan ²⁴	1937	195			0.08%	1

^a $u(T)$ and $u(p)$ denote the standard uncertainties of T and p ; u_r denotes the relative standard uncertainty of the property.

longitudinal and transverse sound speeds were measured at the normal sublimation temperature by Hovi and Mäntylä.²⁵ The densities required to obtain κ_S from the ultrasonic data were obtained from Krupskii *et al.*²⁹ with negligible additional uncertainty.

The sublimation pressure of CO₂ has been measured by numerous workers and the data sources considered in this work are summarized in Table 2. Other pre-1950 sources, all of them designated as group 3, have been listed by Span and Wagner.⁵ The recommendations of Span and Wagner have been followed in selecting three datasets for group 1.^{34,37,38} These extend over the range from 154 K to the triple point. The enthalpy of sublimation $\Delta_s^g H_m$ was measured at the normal sublimation point by Giauque and Egan, with a relative uncertainty of 0.08%, and by Eucken and Donath³¹ at $T = 140$ and 170 K with a relative uncertainty of 0.2%; both sources were assigned to group 1. Maass and Barnes¹⁹ also reported a value of $\Delta_s^g H_m$ at the normal sublimation temperature, but this lacks an uncertainty estimate and was assigned to group 2. The enthalpy of melting, $\Delta_s^l H_m$, at the triple point is reported in three publications^{19,20,40} and the results, which span a range of $\pm 5\%$ about the mean, were assigned to group 2.

4.2. High-pressure data

Properties of solid CO₂ at elevated pressures have been measured at temperatures between 77 and 764 K. The available single-phase data are summarized in Table 3,^{17,41–46} while phase-change data on the melting curve are summarized in Table 4.^{8,19,20,40,47–52} Four sets of molar volume data at $295 \leq T/K \leq 300$ have been reported at pressures of up to 12 GPa;^{17,43,44,46} all were obtained by x-ray diffraction measurements on samples of solid CO₂ confined within a diamond anvil cell. The study of Liu extends to 50 GPa,⁴⁴ but only the data at $p \leq 11$ GPa were used in the fitting.

Shimizu *et al.*⁴⁵ obtained the isentropic bulk modulus $K_S = 1/\kappa_S$ by means of Brillouin scattering measurements on a single crystal of solid CO₂ grown within a diamond anvil cell. The interpretation of these data is complicated slightly by the high frequencies of 5 to 10 GHz associated with the Brillouin scattering measurements. Given the long relaxation time of the internal vibrational modes of CO₂ at $T = 300$ K,

the effective heat capacity of the solids at high frequencies must be that associated with the lattice modes only. Thus, when comparing with the equation of state, the isentropic bulk modulus was computed from the isothermal bulk modulus K_T according to the relation,

$$K_S = (C_{p,m} - C_{int})K_T / (C_{v,m} - C_{int}), \quad (1)$$

where C_{int} is the contribution of the internal vibrational modes of CO₂ to the molar heat capacity of the solid.

Unfortunately, as discussed by Olinger,⁴³ the low-temperature compression data of Stevenson⁴² appear to be in error, and the “rough” compression data of Bridgman at $T = 194$ K are too imprecise to use,⁴¹ leaving the study of Giordano *et al.*¹⁷ as the only one in the single-phase region extending to temperatures outside the narrow range 295 to 300 K.

The only melting-pressure data included in group 1 are those of Clusius *et al.*,⁴⁹ which are confined to pressures below 25 MPa. Other high-quality data sets exist and these extend to much higher pressures. However, plausible uncertainties in the equation of state of the fluid phase significantly exceed the experimental uncertainty of these high-pressure melting-curve data so that, if used in parameter optimization, systematic errors would propagate from the fluid-phase equation.

4.3. Adjustment and weighting of the data

As far as possible, the International Temperature Scale of 1990 (ITS-90)^{53,54} has been used in this work. Accordingly, where feasible, the temperatures given in the work published prior to 1990 have been converted from the scale on which they were measured to ITS-90. Temperatures reported on the International Practical Temperature Scale of 1968 (IPTS-68) were converted to ITS-90 according to the procedure described by Rusby.⁵⁵ Temperatures reported on the International Practical Temperature Scales of 1948 (IPTS-48) and 1927 (ITS-27), which are identical above 90 K, were first converted to IPTS-68 using the algorithm described by Bedford and Kirby⁵⁶ then thence to ITS-90. These corrections are only significant in relation to sublimation pressure and melting-pressure data, as other properties vary sufficiently

TABLE 3. Summary of high-pressure single-phase data

Property	Source	Year	T/K	$p_{\max}/\text{MPa}$	u_r^a	Group
K_S	Shimizu <i>et al.</i> ^{45c}	1993	300	6060	5% ^b	2
V_m	Olinger ⁴³	1982	296	10 110	1% ^b	1
V_m	Liu ⁴⁴	1984	296	10 900	0.6%	1
V_m	Giordano <i>et al.</i> ¹⁷	2010	295 to 764	12 550	0.6%	1
V_m	Scott <i>et al.</i> ⁴⁶	2010	300	9200	0.7% ^b	2
V_m	Stevenson ^{42c}	1957	77 to 185	1000		3
V_m	Bridgman ⁴¹	1938	194	3431		3

^a u_r denotes relative standard uncertainty.

^bEstimated.

^cData reported graphically only.

TABLE 4. Summary of phase-change data on the melting curve

Property	Source	Year	T/K	$u(T)^a/\text{K}$	$u(p)^a/\text{MPa}$	u_r^a	Group
$\Delta_s^l V_m$	Bridgman ⁴⁷	1914	268 to 367			5% ^b	2
$\Delta_s^l H_m$	Kuenen and Robson ⁴⁰	1902	216.6				2
$\Delta_s^l H_m$	Maass and Barnes ¹⁹	1926	216.6				2
$\Delta_s^g H_m$	Eucken and Hauck ²⁰	1928	216.6				2
p_{melt}	Bridgman ⁴⁷	1914	268 to 367	0.1		0.1%	2
p_{melt}	Michels <i>et al.</i> ⁴⁸	1942	217 to 266	0.02		0.1%	2
p_{melt}	Clusius <i>et al.</i> ⁴⁹	1960	217 to 222	0.02		0.1%	1
p_{melt}	Grace and Kennedy ^{50c}	1967	217 to 425	3	50		3
p_{melt}	Iota and Yoo ⁵¹	2001	640	5	1000		3
p_{melt}	Giordano <i>et al.</i> ^{8c,d}	2006	297 to 811	5	20 to 100 ^e		2
p_{melt}	Abramson and Brown ⁵²	2007	295 to 620	1	20 to 40 ^e	2%	2

^a $u(T)$ and $u(p)$ denote the standard uncertainties of T and p ; u_r denotes the relative standard uncertainty of the property.

^bEstimated.

^cData reported graphically only.

^dNumerical data provided by the original authors.

^eIncreases linearly over the temperature range.

slowly with temperature for small differences between practical thermometry scales to be unimportant.

Amongst the group 1 data, there are several cases in which a straightforward scale conversion could not be made. Manzhelii *et al.*²⁶ give no information relating to the temperature scale used between 2 and 190 K and no corrections were made. Giauque and Egan²⁴ used a “thermodynamic scale” between 15 and 190 K, the defining point of which they took to be $T = 273.10$ K at the ice point. Accordingly, their temperature values were scaled by the factor $273.15/273.10$. The melting temperatures of Michels *et al.*⁴⁸ and Clusius *et al.*⁴⁹ were both measured with thermometers calibrated at the temperature of the triple point of CO_2 making use of values for that temperature that are not in agreement with the present value. Therefore, the procedure adopted by Span and Wagner⁵ was followed whereby the temperatures reported by Michels *et al.*⁴⁸ were corrected by subtracting 0.04 K after conversion to ITS-90; similarly those of Clusius *et al.*⁴⁹ were lowered by 0.05 K after conversion. In group 2, the melting-curve data of Bridgman⁴⁷ pre-date any agreed international scale. However, Bridgman reported temperatures in degree Celsius and, as all were within the range 0 to 100 °C, these Celsius temperatures were simply multiplied by 0.99974, to account for the steam point being 99.974 °C on ITS-90, before conversion to Kelvin. This correction is of marginal significance.

Calorimetric data reported in calories were converted to SI units by assuming that 1 cal = 4.184 J, except in one case²⁴ where the authors gave a definition of their “15° calorie” as 4.185 J.

The fitted parameters in the equation of state model were found by minimization of the objective function,

$$\chi^2 = \frac{\sum_{i=1}^N w_i [(X_i - X_{i,\text{fit}})/u(X_i)]^2}{\sum_{i=1}^N w_i} \quad (2)$$

Here, X is any thermodynamic property, X_i is the i th experimental datum, $u(X_i)$ is the standard uncertainty of X_i , $X_{i,\text{fit}}$ is the value of property X at state point i computed from the equation of state, w_i is a weighting factor, and N is the total number of data points considered. This dimensionless statistic is defined such that $\chi^2 \rightarrow 1$ for a “good” fit.

The primary weighting of the experimental data was provided by the standard uncertainties $u(X_i)$, which were known or estimated for all fitted data, and hence normally $w_i = 1$. However, in order to compensate for the uneven distribution of the data in terms of both number and property, a few data were assigned weights other than unity. These cases are detailed specifically in Sec. 6.2.

All types of primary data were included simultaneously in the fit. In the case of sublimation- and melting-pressure data, the fitted quantity was the molar Gibbs energy G_m . This quantity was computed for the fluid phases from the equation of state of Span and Wagner.⁵ The standard uncertainty $u(G_m)$ ascribed to these data was computed from the standard uncertainties in temperature and pressure according to the equation:

$$[u(G_m)]^2 = [S_m u(T)]^2 + [V_m u(p)]^2, \quad (3)$$

where S_m and V_m are the molar entropy and the molar volume of the fluid phase.

4.4. Fluid equation of state

In order to carry out solid–fluid phase-equilibrium calculations, the difference between the molar Gibbs energies of the solid and fluid phases is required as a function of temperature and pressure; phase equilibrium states being those in which this difference vanishes. The reference equation of state for fluid CO_2 , that of Span and Wagner,⁵ has an arbitrarily assigned reference state in which the entropy and enthalpy of the hypothetical perfect gas are set to zero. Since only differences in enthalpy or entropy between equilibrium states can

be observed experimentally, the reference state and the values assigned for the enthalpy and entropy may be changed without affecting any observable property. When the solid phase is included, it is of course essential that the choice of reference state and reference values of enthalpy and entropy for the fluid are consistent with the equation of state for the solid. In this work, the molar Gibbs energies of the solid and fluid phases were made exactly equal at the triple-point temperature and pressure, given by Span and Wagner as $T_t = 216.592$ K and $p_t = 0.51795$ MPa. This is achieved by making the triple point the reference state for the fluid equation of state and choosing appropriate reference values for the molar entropy and molar enthalpy of the gas phase in that state.

The reference value for the molar entropy of the gas phase at the triple point $S_m(g, T_t, p_t)$ was separated into a perfect-gas part, which was evaluated from statistical mechanics, and a residual part, which was obtained from the Span–Wagner equation of state.⁵ The statistical entropy of the perfect gas was evaluated as follows:

$$S_m^{\text{pg}}(T, p) = S_{\text{tr}}(T, p) + S_{\text{rr}}(T) + S_{\text{int}}(T). \quad (4)$$

Here, S_{tr} is the contribution of the translational modes given by

$$S_{\text{tr}}(T, p) = R \ln \left[(2\pi m k_B T / h^2)^{3/2} (k_B T / p) e^{5/2} \right], \quad (5)$$

where m is the mass of one molecule, k_B is Boltzmann's constant, and h is Planck's constant. The term S_{rr} is the rigid-rotor approximation for the rotational molar entropy,

$$S_{\text{rr}}(T) = R \ln \{ eT / (\sigma \theta_{\text{rot}}) \}, \quad (6)$$

in which $\sigma = 2$ is the symmetry number, $\theta_{\text{rot}} = hcB_0/k_B$ is the rotational characteristic temperature, and B_0 is the rotational constant of the molecule in the ground vibrational state. The final term, S_{int} , combines the molar entropy associated with internal vibrations (including all anharmonic effects) with the corrections required to account for vibration–rotation interaction and centrifugal distortion. This term was calculated from the model of Span and Wagner as follows:⁵

$$S_{\text{int}}(T) = R \sum_{i=1}^5 C_i \left[(\varphi_i \tau) \{ \exp(\varphi_i \tau) - 1 \}^{-1} - \ln \{ 1 - \exp(-\varphi_i \tau) \} \right], \quad (7)$$

where $\tau = T_c / T$, $T_c = 304.1282$ K is the critical temperature of CO₂ and the coefficients C_i and φ_i are given in Table 5.

Natural carbon dioxide is a mixture of isotopes and, in principle, one should include both the entropy of mixing and the effects of nuclear-spin statistics. However, this has not been done in the present case as the solid-phase equation of state also neglects these terms. To obtain the relevant mean molecular constants required to compute S_m^{pg} for the natural isotopic mixture, we consider the five most abundant isotopic species each having mass m_i and ground-state rotational constants $B_{0,i}$ in terms of which

$$\ln m = \sum_i A_i \ln m_i, \quad (8)$$

and

$$\ln B_0 = \sum_i A_i \ln B_{0,i}. \quad (9)$$

The atomic masses were taken from Wieser and Berglund,⁵⁷ the abundances from Rosman and Taylor,⁵⁸ and the rotational constants from Chedin.⁵⁹ Equations (8) and (9) evaluated to give $m = 7.30790 \times 10^{-26}$ kg, $B_0 = 0.39012$ cm⁻¹ and hence $\theta_{\text{rot}} = 0.56129$ K. With these values of the molecular constants, the perfect-gas statistical molar entropy of CO₂ at its triple point was found to be 22.7200R. The calculation method was compared with the results of Woolley⁶⁰ at a nearby state point ($T = 220$ K, $p = 0.101325$ MPa), and the present method was found to provide a value in agreement to within 0.0001R. From the Span–Wagner equation of state, the residual entropy of the gas phase at the triple point is $-0.1959R$, so that the final calculated value of the “absolute” molar entropy of the gas in the reference state is

$$S_m(g, T_t, p_t) / R = 22.5242. \quad (10)$$

The molar enthalpy of the gas in the reference state, $H_m(g, T_t, p_t)$, was adjusted such that $H_m(g, T_t, p_t) - TS_m(g, T_t, p_t)$ was exactly equal to the value of the molar Gibbs energy of the solid at the same temperature and pressure obtained from the new equation of state. This led finally to

$$H_m(g, T_t, p_t) / RT_t = 17.9287. \quad (11)$$

The equation of state of Span and Wagner is valid for $p \leq 800$ MPa.⁵ In comparing group 2 phase-change data for

TABLE 5. Parameters required for calculation of the statistical entropy of the perfect gas

Species	A_i	$M_i / (\text{kg mol}^{-1})$	$B_{0,i} / \text{cm}^{-1}$	i	C_i	φ_i
¹² C ¹⁶ O ₂	0.98453	0.043990	0.39022	1	1.99427042	3.15163
¹² C ¹⁶ O ¹⁷ O	0.00075	0.044994	0.37862	2	0.62105248	6.11190
¹² C ¹⁶ O ¹⁸ O	0.00403	0.045994	0.36819	3	0.41195293	6.77708
¹³ C ¹⁶ O ₂	0.01065	0.044993	0.39024	4	1.04028922	11.32384
¹³ C ¹⁶ O ¹⁸ O	0.00004	0.046997	0.36818	5	0.08327678	27.08792

solid CO₂ with the new equation of state at pressures above 800 MPa, calculations involving extrapolation of the Span–Wagner model have been considered. Clearly, the results need to be treated with caution. A new equation of state for fluid CO₂ at $p \geq 250$ MPa and $300 \leq T/K \leq 700$ was proposed by Giordano *et al.*⁸ This model was developed by thermodynamic integration of sound-speed data obtained from Brillouin scattering experiments, starting from initial values of density and heat capacity computed from the Span–Wagner equation on the isobar $p = 250$ MPa. The results were fitted to a 12-parameter function expressing the density as a function of T and p . This model has also been used for comparisons with group 2 data that call for thermodynamic properties of the fluid. When using the Giordano equation of state in this work, the entropy and enthalpy were matched to those given by the Span–Wagner equation at the temperature in question and at $p = 800$ MPa; changes in these quantities with pressure were obtained by isothermal numerical integration of the appropriate derivatives of the Giordano equation.

5. Thermodynamic Modeling Basis

Given the sparse data set available, a theoretically based equation of state was favored containing a relatively small number of parameters, most of which have a physical interpretation and hence some range of expected values. The quasi-harmonic approximation has been widely used to obtain the equation of state of inorganic crystalline solids and has been applied previously to solid CO₂.¹⁷ This approximation was felt to be a useful basis for establishing the thermodynamic model in this work.

5.1. The quasi-harmonic approximation

The essence of the quasi-harmonic approximation (QHA) is to represent the lattice vibrations of a crystalline solid by means of a set of independent harmonic oscillators, the frequencies of which are functions of the lattice volume only. In this approximation, the configurational Helmholtz energy A for a crystal of volume V containing N molecules at temperature T is given by

$$A(N, V, T) = \sum_{i < j} u_{ij}(V/N) + \frac{1}{2} \sum_{\mathbf{k}, l} \hbar \omega_l(\mathbf{k}, V/N) + k_B T \sum_{\mathbf{k}, l} \ln[1 - \exp\{-\hbar \omega_l(\mathbf{k}, V/N)/k_B T\}], \quad (12)$$

where $\omega_l(\mathbf{k}, V/N)$ is the angular frequency of the l th harmonic lattice mode for phonons of wavevector $\mathbf{k}$, and $u_{ij}(V/N)$ is the effective pair-potential-energy function for the system which, for a given crystal structure, is a function of V/N .

It is usual to combine the zero-point lattice vibrational energy and the intermolecular potential energy in a single term, $U_0(N, V)$, that gives the total configurational internal energy of the system at zero temperature. The sum over the external lattice vibrations, the third term in Eq. (12), is then

expressed in terms of the density of state $G(\omega, V/N)$ defined such that

$$A(N, V, T) = U_0(N, V) + Nk_B T(m/n) \times \int_0^\infty G(\omega, V/N) \ln[1 - \exp\{-\hbar \omega/k_B T\}] d\omega. \quad (13)$$

Here, m is the number of external degrees of freedom per unit cell, n is the number of molecules in the unit cell, and $G(\omega, V/N)$ is normalized such that

$$\int_0^\infty G(\omega, V/N) d\omega = 1. \quad (14)$$

For solid phase I of CO₂, there are 4 molecules per unit cell and 20 external degrees of freedom, of which 12 correspond to pseudo-translational vibrations and the remaining 8 correspond to pseudo-rotational librations.⁶¹ Thus, $m = 20$ and $n = 4$. Each molecule possesses a further 4 internal vibrational modes, so the total number of internal and external degrees of freedom per unit cell is 36.

The 20 external degrees of freedom comprise 3 acoustic modes and 17 optical modes. The frequencies of these optical modes at zero wavevector are known from a combination of infrared and Raman spectroscopy^{59,62–68} and inelastic neutron scattering.²⁸ However, the only information available for the acoustic modes and for the optical modes at *non-zero* wavevector come from a single study by inelastic neutron scattering.²⁸ That study was restricted to just two high-symmetry planes in the CO₂ crystal, and the information is insufficient to construct the full density of states; thus some further simplification is essential.

5.2. Debye–Einstein approximation

In order to proceed to a convenient equation of state, a Debye–Einstein approximation is adopted in which the density of states is represented by the combination of a Debye distribution, accounting for 3/5 of the total weight, and a sum of discrete harmonic terms, accounting for the remaining 2/5 of the total weight. The former represent the vibrons, while the latter represent the librons. The expression for G is

$$G = \frac{9\omega_D^2}{5\omega_D^3} H(\omega_D - \omega) + \sum_{i=1}^m a_i \delta(\omega_i - \omega), \quad (15)$$

where ω_D is the angular cut-off frequency for the Debye distribution, $H(x)$ is the unit step function,^{b)} $\delta(x)$ is the Dirac delta function, m is the number of discrete harmonic modes, ω_i is the angular frequency (and a_i the weight) of the i th discrete harmonic term, and $\sum_{i=1}^m a_i = 2/5$. It follows that the total molar Helmholtz energy $A_m(T, V_m)$ at temperature T and molar volume V_m is

^{b)} $H(x) = 0$ for $x < 0$ and $H(x) = 1$ for $x \geq 0$.

$$A_m(T, V_m) = A_0(V_m) + A_D(T, V_m) + \sum_{i=1}^m a_i A_i(T, V_m) + A_{\text{int}}(T). \quad (16)$$

The terms represented here arise as follows: A_0 from the intermolecular and zero-point lattice energy of the crystal at $T = 0$ K; A_D from the Debye distribution of lattice vibrations; A_i from the i th discrete harmonic component of the lattice vibrations; and A_{int} from the internal molecular vibrations. All of the thermodynamic properties may be derived from this Helmholtz energy function by differentiation.

5.3. Grüneisen parameters

In the following, it is convenient to introduce the characteristic temperatures $\theta_D = (\hbar\omega_D/k_B)$ and $\theta_i = (\hbar\omega_i/k_B)$ which, like ω_D and ω_i , are functions of the molar volume of the crystal. This dependency is expressed in terms of the Grüneisen parameters γ_D and γ_i , which are defined by the following relations:

$$\left. \begin{aligned} \gamma_D &= -(\mathrm{d} \ln \theta_D / \mathrm{d} \ln V_m) \\ \gamma_i &= -(\mathrm{d} \ln \theta_i / \mathrm{d} \ln V_m) \end{aligned} \right\}. \quad (17)$$

Experimentally, Grüneisen parameters are found to vary somewhat with expansion or compression of the crystal, and this may be represented empirically by means of the following model:

$$\left. \begin{aligned} \gamma_D &= \gamma_{D,0} (V_m/V_{00})^{q_D} \\ \gamma_i &= \gamma_{i,0} (V_m/V_{00})^{q_i} \end{aligned} \right\}. \quad (18)$$

Here V_{00} is the molar volume of the crystal at zero temperature and pressure, $\gamma_{D,0}$ and $\gamma_{i,0}$ are the Grüneisen parameters in that state, and q_D , q_i are exponents to be adjusted for each mode. For non-zero exponent values, this model leads to the following explicit expressions for the characteristic temperatures at molar volume V_m ,

$$\left. \begin{aligned} \theta_D &= \theta_{D,0} \exp[(\gamma_{D,0}/q_D)\{1 - (V_m/V_{00})^{q_D}\}] \\ \theta_i &= \theta_{i,0} \exp[(\gamma_{i,0}/q_i)\{1 - (V_m/V_{00})^{q_i}\}] \end{aligned} \right\}. \quad (19)$$

If $q_i = 0$, then $\theta_i = \theta_{i,0} (V_m/V_{00})^{-\gamma_i}$.

5.4. The cold curve

Numerous equations of state have been proposed to represent the cold compression and expansion behavior of atomic and molecular crystals. Many of these are based on model intermolecular potential-energy functions or on consideration of generalized stress-strain relations. In the present work, the fourth-order logarithmic equation of state⁶⁹ was adopted because of its simple form and proven ability to represent extreme compression behavior with high accuracy. The molar Helmholtz energy for this model may be expressed in terms of the reduced density $z = (V_{00}/V_m)$ as follows:

$$A_0 = V_{00} \left[\frac{1}{2} c_1 \ln^2 z + \frac{1}{3} c_2 \ln^3 z + \frac{1}{4} c_3 \ln^4 z \right], \quad (20)$$

where c_1 , c_2 , and c_3 are parameters related to the bulk modulus of the cold crystal and its first two derivatives with respect to pressure.⁷⁰ The contribution of this term to the pressure and its first derivative, are obtained by differentiation of A_0 with respect to V_m with the following result:

$$p_0 = c_1 z \ln z + c_2 z \ln^2 z + c_3 z \ln^3 z, \quad (21)$$

$$\left(\mathrm{d}p_0 / \mathrm{d}V_m \right) = -(z/V_m) [c_1(1 + \ln z) + c_2(2 + \ln z) \ln z + c_3(3 + \ln z) \ln^2 z]. \quad (22)$$

5.5. External modes

The molar Helmholtz energy for the Debye distribution of harmonic oscillators is given by

$$A_D = RT[3 \ln\{1 - \exp(-\theta_D/T)\} - D_3(\theta_D/T)], \quad (23)$$

where $D_3(z)$ is the Debye function,

$$D_3(z) = (3/z^3) \int_0^z \frac{x^3 \mathrm{d}x}{e^x - 1}. \quad (24)$$

Differentiation of Eq. (23) leads to the contributions to the molar entropy S_m , molar internal energy U_m , isochoric molar heat capacity $C_{V,m}$, pressure p and derivatives as follows:

$$S_D = R[4D_3(\theta_D/T) - 3 \ln\{1 - \exp(-\theta_D/T)\}], \quad (25)$$

$$U_D = 3RTD_3(\theta_D/T), \quad (26)$$

$$C_{V,D} = 3R[D_3(\theta_D/T) - (\theta_D/T)D'_3(\theta_D/T)], \quad (27)$$

$$p_D = (\gamma_D/V_m)U_D, \quad (28)$$

$$\left(\partial p_D / \partial V_m \right)_T = (\gamma'_D - \gamma_D/V_m)(p/\gamma_D) - R\theta_D(\gamma_D/V_m)^2 D'_3(\theta_D/T), \quad (29)$$

$$\left(\partial p_D / \partial T \right)_V = (\gamma_D/V_m)C_{V,D}. \quad (30)$$

Here, $D'_3(z)$ is the first derivative of $D_3(z)$,⁷¹ and the prime denotes differentiation with respect to V_m . The subscript “D” indicates that Eqs. (23) and (25) to (30) represent the contribution of the Debye term to A_m , S_m , U_m , $C_{V,m}$, p , and its derivatives, respectively. Equation (23) is normalized such that the Debye contribution to the isochoric molar heat capacity, $C_{V,D}$, approaches $3R$ as $T \rightarrow \infty$.

Numerical evaluation of the Debye function $D_3(z)$ may be accomplished to machine accuracy either by numerical quadrature or by means of the efficient expansion in Chebyshev polynomials described by Macleod.⁷¹

⁷⁰This is given by $D'_3(z) = -(3/z)D_3(z) + 3/(e^z - 1)$.

For the discrete harmonic modes, the molar Helmholtz energy term A_i is given by

$$A_i = RT \ln\{1 - \exp(-\theta_i/T)\}. \quad (31)$$

Each such term is associated with the following contributions to S_m , U_m , $C_{V,m}$, p and its derivatives:

$$S_i = R[(\theta_i/T)/\{\exp(\theta_i/T) - 1\} - \ln\{1 - \exp(-\theta_i/T)\}], \quad (32)$$

$$U_i = R\theta/\{\exp(\theta_i/T) - 1\}, \quad (33)$$

$$C_{V,i} = R(\theta_i/T)^2 \exp(\theta_i/T)/\{\exp(\theta_i/T) - 1\}^2, \quad (34)$$

$$p_i = (\gamma_i/V_m)U_i, \quad (35)$$

$$\begin{aligned} (\partial p_i/\partial V_m)_T &= (p_i\gamma_i/V_m)[(V_m\gamma'_i - \gamma_i)/\gamma_i^2 \\ &\quad + (\theta_i/T) \exp(\theta_i/T)/\{\exp(\theta_i/T) - 1\} - 1], \end{aligned} \quad (36)$$

$$(\partial p_i/\partial T)_{V_m} = (\gamma_i/V_m)C_{V,i}. \quad (37)$$

Equation (31) is normalized such that the derived isochoric molar heat capacity, $C_{V,i}$, of each mode approaches R as $T \rightarrow \infty$. However, the weights a_i which appear in Eq. (15) multiply each A_i and all of the contributions to other thermodynamic properties derived therefrom.

5.6. Internal modes

In this work, the internal vibrations of carbon dioxide molecules in the solid phase are also represented by quasi-harmonic terms, and so Eq. (31) provides the basis for modeling each internal mode. Isolated CO_2 molecules possess three distinct fundamental modes: the singly degenerate symmetric and asymmetric stretching modes and the doubly degenerate bending mode. These same modes have been identified in the infrared and Raman spectrum of solid phase I, but with frequencies that are shifted slightly from the values observed in the gas phase. The most complete study is that of Hanson and Jones,⁶⁵ in which the fundamental modes were determined as a function of pressure up to 12 GPa at $T = 298$ K. In order to determine the values of the corresponding characteristic temperatures at $V_m = V_{00}$ and the associated Grüneisen parameters, the experimental wavenumbers ν_i were analyzed as functions of the molar volume V_m calculated from the isothermal equation of state measured by Giordano *et al.*¹⁷ at almost the same temperature, together with value $V_{00} = 25.80 \text{ cm}^3 \text{ mol}^{-1}$ obtained from Krupskii.²⁹ As shown in Fig. 2, a linear dependence of $\ln(\nu_i)$ upon $\ln(V_m/V_{00})$ was found for each mode from which the Grüneisen parameters and the values of $\theta_{i,0}$ have been determined. These values are given below in Table 6. The Grüneisen parameters for the internal modes are much smaller than those of the external modes, and so the variation of the characteristic temperatures with volume is slight. In this work, it was assumed that this small dependence can be represented

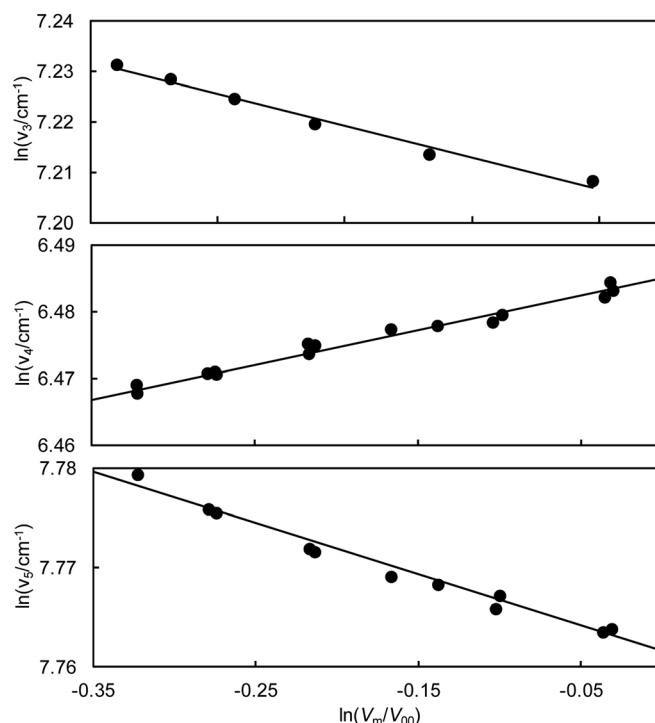


FIG. 2. Wavenumbers ν_i of the internal vibrational modes of solid CO_2 at $T = 298$ K as functions of molar volume: ●, experimental data of Hanson and Jones;⁶⁵ —, linear regression. The modes are designated here $i = 3$ for the symmetric stretch, $i = 4$ for the bending modes, and $i = 5$ for the asymmetric stretch.

at all temperatures with sufficient accuracy using the values of γ_i determined at $T = 298$ K; thus $q_i = 0$ for these modes.

5.7. Anharmonic corrections

Solid CO_2 is known to be an orientationally ordered solid with no low-pressure order–disorder phase transition such as that exhibited by nitrogen. Nevertheless, various deviations from the QHA are expected to be important especially near the triple point, where the crystal is at its most expanded,

TABLE 6. Parameters of the new equation of state

Parameter	Value	Parameter	Value
$V_{00}/(\text{cm}^3 \text{ mol}^{-1})$	25.800	b_1	1.15
c_1/MPa	9000	b_2	55.428
c_2/MPa	32 315	b_3	8.8
c_3/MPa	941	θ_D/K	148.9
θ_1/K	92.3	a_1	0.5
θ_2/K	179.4	a_2	1.5
θ_3/K	1945.9	a_3	1
θ_4/K	942.7	a_4	2
θ_5/K	3379.2	a_5	1
γ_D	2.6163	q_D	−0.21
γ_1	4.0463	q_1	−0.21
γ_2	1.1259	q_2	−0.21
γ_3	0.0633	q_3	0
γ_4	−0.0521	q_4	0
γ_5	0.0515	q_5	0

and at high temperatures. Manzhelii *et al.*²⁶ attributed part of the temperature dependence of the heat capacity along the sublimation curve to the creation of point defects in which isolated molecules lose their orientation. They presented a model specifically for this type of anharmonicity. Landau and Lifshitz⁷² developed a high-temperature perturbation expansion to account for small deviations from harmonic behavior and showed that the leading term in the Helmholtz energy varies like the product of T^2 and some function of molar volume to be determined. Unfortunately, this simple relation is restricted to high temperatures as the resulting contribution to $C_{p,m}$, being linear in T , is unphysical as $T \rightarrow 0$. Wu and Wentzcovitch⁷³ applied a renormalization approach to correct the QHA for anharmonic effects and showed that a single temperature- and volume-independent parameter was sufficient to account for the observed anharmonicity in simple crystalline metal oxides. That work confirmed that the effect of anharmonicity on the Helmholtz energy varies like T^2 at high temperatures but, unlike the perturbation expansion, it rolls over to the correct T^4 behavior at low temperatures. The results presented by Wu and Wentzcovitch show a very simple exponential increase with molar volume and the entire variation of the anharmonic correction A_A to the QHA molar Helmholtz energy can be represented by the following simple expression:

$$A_A = b_1 R [(T/\theta_{D,0})^4 / \{1 + b_2 (T/\theta_{D,0})^2\}] \exp[b_3 \{(V_m - V_{00})/V_{00}\}]. \quad (38)$$

Although this involves three parameters (b_1 , b_2 , b_3) rather than one, it was preferred in this work as it is more direct to apply than the renormalization approach which results in a temperature- and pressure-dependent volume translation. Fortunately, as described below, only two of the parameters in Eq. (38) needed to be optimized in the fit to experimental data.

The inclusion of the anharmonic term implies a departure from the Dulong–Petit limit for the lattice heat capacity at high temperatures. For positive (negative) b_1 , the departure is negative (positive).

6. Parameter Optimization

6.1. Pre-fitting

Having fixed on the general structure of the QHA equation of state, the first question to be addressed was the number of harmonic terms necessary to represent adequately the effect of the librational modes. This was studied by first considering the isochoric molar heat capacity of the crystal along the sublimation curve. For this purpose, $C_{V,m}$ was obtained from the experimental $C_{p,m}$ data^{24,26} as follows:

$$C_{V,m} = C_{p,m} [1 + T\alpha^2 V_m / (C_{p,m} \kappa_S)]^{-1}. \quad (39)$$

For this calculation, α and V_m were obtained from Krupskii *et al.*²⁹ and κ_S from Manzhelii *et al.*²⁶ The low-temperature

data constrain the Debye temperature $\theta_{D,0}$ to values close to 151 K, and it was found that all of the data could be represented within a few percent with the addition of just two harmonic terms of equal weight. In this calculation, the characteristic temperatures were held constant, independent of V_m , and so this was a simple harmonic model. Additional terms, variable weights, and non-zero Grüneisen parameters did not significantly improve the fit to $C_{V,m}$. Of course these initial calculations are subject to the effects of uncertainties in the other quantities appearing in Eq. (39), especially those in α and κ_S . Nevertheless, the results were used to constrain the number of terms used for the librational modes.

As $T \rightarrow 0$ K along the sublimation curve, the thermal Grüneisen parameter, defined by

$$\gamma_{th} = \alpha V_m K_T / C_{V,m} \quad (40)$$

is expected to approach a constant value which, in the present Debye–Einstein approximation, equals the Grüneisen parameter of the Debye distribution evaluated at $V_m = V_{00}$. Furthermore, as $T \rightarrow 0$ K along the sublimation curve, the isothermal bulk modulus K_T approaches the value of parameter c_1 in the equation for the cold compression curve, and $C_{V,m}$ approaches the Debye limit,

$$C_{V,m} = (12\pi^4 R/5)(T/\theta_D)^3. \quad (41)$$

It follows that αV_m should vary like T^3 in the approach to that limit, so that

$$V_m = V_{00} + b(T/\theta_D)^4 \quad (42)$$

and

$$\gamma_{D,0} = 20(b/\theta_D)c_1/(12\pi^4 R). \quad (43)$$

Figure 3 shows the low-temperature variation of V_m , which does indeed conform to Eq. (42) and yields $b = (21.4 \pm 0.7) \text{ cm}^3 \text{ mol}^{-1}$ with $\theta_D = 151.5$ K.

The pre-fitting described in this section was used only to determine: (a) that two quasi-harmonic modes were required

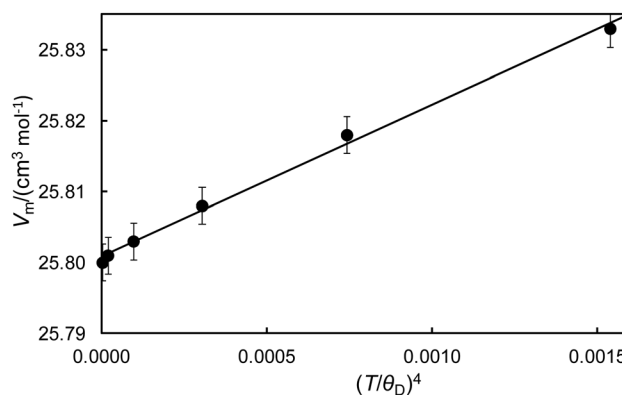


Fig. 3. Molar volume V_m of solid CO₂ as a function of $(T/\theta_D)^4$ with $\theta_D = 151.5$ K, for $T \leq 30$ K: ●, experimental data from Krupskii *et al.*;²⁹ —, linear regression line.

for the librational modes and (b) that $(\gamma_{D,0}/c_1)$ should be constrained to the value $(2.91 \pm 0.03) \times 10^{-4} \text{ MPa}^{-1}$ implied by Eq. (43). In the main fitting, the effects of relaxing either or both of these constraints were checked.

6.2. Global fit

The equation of state as finally implemented is given by

$$A_m(T, V_m) = A_0(V_m) + A_D(T, V_m) + \sum_{i=1}^5 a_i A_i(T, V_m) + A_A(T, V_m), \quad (44)$$

where A_0 is given by Eq. (20), A_D by Eq. (23), A_i by Eq. (31), and A_A by Eq. (38). The sum over harmonic terms extends to $i=5$ to include two librational modes and three internal modes. The characteristic temperatures θ_D and θ_i are given by Eq. (19) (or by the appropriate limit when $q_i \rightarrow 0$). There were thus a total of 30 parameters in the model. However, the number of parameters to be adjusted in the optimization was immediately reduced by independent data and by constraints to a maximum of 16. There were 12 independently fixed parameters, being the values of a_i , θ_i , γ_i , and q_i for the internal modes $i=3$ (symmetric stretch), $i=4$ (bending modes), and $i=5$ (asymmetric stretch). In addition, there were two further constraints: $a_1 + a_2 = 2$ and the use of Eq. (43) for $\gamma_{D,0}$.

Initial values of the remaining parameters were $\theta_D = 151.5 \text{ K}$, $\theta_1 = 100 \text{ K}$, $\theta_2 = 200 \text{ K}$, $a_1 = a_2 = 1$ (all estimated from pre-fitting the heat capacity data), $\gamma_D = \gamma_1 = \gamma_2 = 2.1$ from Manzhelii *et al.*,²⁶ $q_D = q_1 = q_2 = 0$, and $b_1 = 0$ (so that the anharmonic term was absent). Fitting all 13 free parameters that remained after eliminating the anharmonic term proved to be difficult in the sense that some parameters were highly correlated and, in particular, unconstrained minimization of χ^2 tended to push the exponents q_D , q_1 , q_2 to extreme values. However, it was found that including the additional constraints $q_D = q_1 = q_2$ resulted in a much more robust model. It was also found that the optimization drove a_1 to a value very close to 0.5; accordingly, it was constrained to exactly that value in subsequent analysis. These constraints reduced the number of free parameters to 10 and, upon optimization, resulted in values that were not affected by plausible changes in the initial values. Nevertheless, the resulting model was somewhat imperfect as systematic deviations remained in both the isobaric heat capacity near the normal sublimation temperature²⁴ and in the high-temperature molar volumes of Giordano *et al.*¹⁷ The molar volume along the sublimation curve was very well fitted in all cases, but those data received very high weighting due to their low relative standard uncertainty of 0.01%. It was possible to accommodate either the $C_{p,m}$ data or the high-temperature V_m data by increasing the weight on one data set or the other, but it was not possible to fit them both within the claimed experimental uncertainties. The high-temperature V_m data favored a larger value of c_1 , in the region of 10 000 MPa, whereas good fits without those data implied c_1 of about 8500 MPa.

The latter is more compatible with the value of the isothermal bulk modulus on the sublimation curve at $T \rightarrow 0 \text{ K}$, equal to c_1 in the model, deduced from the ultrasonic data of Manzhelii *et al.*²⁶ but not included in the fit.

The marginal failure of the quasi-harmonic model alone to reconcile all of the primary data, and especially the deviations of $C_{p,m}$ on the sublimation curve in the approach to the triple point, suggested that anharmonic terms may be significant. Accordingly, the term given by Eq. (38) was included in the analysis. It was found that the parameter b_2 had very little influence on the quality of the fit, and so this was constrained to the value $(\theta_{D,0}/20 \text{ K})^2$ so that the transition to T^4 behavior of A_A occurred for temperatures near 20 K where $C_{V,m}$ is approaching the Debye limit. In the optimization, now involving 12 free parameters, the weighting on the heat capacity data at $T \geq 100 \text{ K}$ and on the molar volume data of Giordano *et al.* was increased by setting $w_i = 5$ for all these points. The quality of the resulting fit was good, as judged by both statistical measures and by the absence of systematic deviations for any property. Finally, with negligible change in χ^2 , the parameters were rounded to an appropriate number of significant figures to give the values presented in Table 6.

7. Comparison with Primary and Secondary Data

The fit as a whole is characterized by $\chi^2 = 3.4$, indicating that the data are, on average, fit largely within the associated uncertainties. The quality of the fit for the individual primary data sets is detailed in Table 7 in terms of χ^2 (calculated for each data set alone), the absolute average relative deviation Δ_{AAD} and the average relative deviation, or bias, Δ_{Bias} . The latter two statistical measures are defined as follows:

$$\Delta_{AAD} = N_X^{-1} \sum_{i=1}^{N_X} |X_i/X_{i,fit} - 1| \quad (45)$$

and

$$\Delta_{Bias} = N_X^{-1} \sum_{i=1}^{N_X} (X_i/X_{i,fit} - 1), \quad (46)$$

where X_i denotes the value of an experimental property, $X_{i,fit}$ is the value of the property determined from the equation of state at the same state point, and N_X is the number of points in the data set for property X under consideration.

Figure 4 shows deviations of the high-pressure molar volumes from the new equation of state. The primary data sets of Giordano *et al.*,¹⁷ Olinger,⁴³ and Liu⁴⁴ are each represented with $\chi^2 \approx 1$, indicating that the model is in excellent agreement with the data. $\Delta_{AAD} = 0.7\%$ for the three data sets combined. Figure 4 shows that the secondary data of Scott⁴⁶ are also represented essentially within their uncertainty. The data for molar volume along the sublimation curve are shown in Fig. 5. The deviations for the primary data of Krupskii *et al.*²⁹ vary from -0.017% to $+0.015\%$, with zero bias, $\chi^2 = 0.8$, and $\Delta_{AAD} = 0.008\%$. The secondary data of Maass and Barnes,¹⁹ Heuse,²¹ and Keesom and Kohler,^{22,23} and

TABLE 7. Absolute average relative deviations Δ_{AAD} and average relative deviations (bias) Δ_{Bias} of the primary data from the new equation of state

Property	T/K	p/MPa	χ^2	Δ_{AAD}	Δ_{Bias}	Data set
V_{m}	295 to 764	$< 12\,300$	1.4	0.6%	0.1%	Giordano <i>et al.</i> ¹⁷
V_{m}	296	$< 10\,100$	0.9	0.7%	0.1%	Olinger ⁴³
V_{m}	296	$< 10\,900$	0.6	0.6%	0.0%	Liu ⁴⁴
V_{m}	6 to 216	< 0.52	0.8	0.008%	0.000%	Krupskii <i>et al.</i> ²⁹
$C_{p,\text{m}}$	2 to 38	≈ 0	7.5	2.1%	0.0%	Manzhelii <i>et al.</i> ²⁶
$C_{p,\text{m}}$	15 to 195	≤ 0.1	2.1	0.4%	0.2%	Giauque and Egan ²⁴
α	13 to 139	≤ 0.1	6.1	11.7%	-9.9%	Manzhelii <i>et al.</i> ²⁶
$\Delta_{\text{s}}^{\text{g}}H_{\text{m}}$	195	0.1	0.6	0.05%	-0.05%	Giauque and Egan ²⁴
$\Delta_{\text{s}}^{\text{g}}H_{\text{m}}$	140 to 170	≤ 0.1	0.6	0.12%	0.12%	Eucken and Donath ³¹
p_{sub}	178 to 198	≤ 0.13	6.6	0.18%	-0.18%	Ambrose ³⁴
p_{sub}	194 to 216	≤ 0.52	3.3	0.06%	0.00%	Fernandez-Fassnacht and del Rio ³⁸
p_{sub}	154 to 216	≤ 0.52	4.3	0.14%	0.09%	Bilkadi <i>et al.</i> ³⁷
p_{melt}	217 to 222	≤ 24	1.0	0.7%	-1.2%	Clusius <i>et al.</i> ⁴⁹

Curzon²⁷ are also shown in Fig. 5. Although the deviations are larger, the agreement is also satisfactory in the light of the probable experimental uncertainties. Thus, for molar volume, there is good agreement with the experimental data at temperatures from 6 K to 764 K.

Figure 6 compares the change in molar volume on melting with the predictions obtained from the new equation of state for the solid together with the equation of Span and Wagner⁵ for the fluid phase, the latter extrapolated for pressures above 800 MPa. The experimental value at the triple point is derived from the solid-phase data of Krupskii *et al.*²⁹ and the fluid-state equation of state; a value at 295 K is similarly derived from the solid-phase data of Giordano *et al.*¹⁷ extrapolated to the melting pressure. The remaining data are the directly measured values of Bridgman.⁴⁷ There is good agreement with the combined solid and fluid models at all pressures, although the model values are systematically slightly higher. This does not necessarily indicate an error in the solid-phase model. Indeed, an alternative calculation using the fluid-phase equation of state of Giordano *et al.*⁸ in place of that of Span and Wagner⁵ gives almost perfect agreement with experiment at $p \geq 500$ MPa.

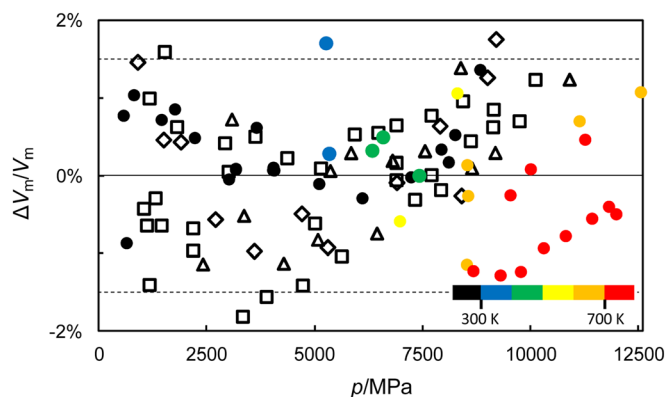


FIG. 4. (Color online) Fractional deviations $\Delta V_{\text{m}}/V_{\text{m}}$ of the molar volume of solid CO₂ from the new equation of state as a function of pressure p at various temperatures T : ●, Giordano *et al.*¹⁷ □, Olinger⁴³ △, Liu⁴⁴ ◇, Scott *et al.*⁴⁶ ---, uncertainty band for the equation of state.

Figure 7 shows the deviations of the experimental isobaric molar heat capacity^{24,26} from the predictions of the equation of state. The quality of the fit to the data of Giauque and Egan²⁴ is good and there is an absence of systematic trends. The worst deviations are found near $T = 25$ K, where several points deviate by more than twice the uncertainty of the data. This may indicate some weakness in the Debye approximation which, at this temperature, dominates the rapidly changing heat capacity. The data of Manzhelii *et al.*²⁶ are also represented without systematic trends, but there is more scatter. It seems that the uncertainties claimed by these authors are too optimistic, as $\chi^2 = 7.5$ for these data yet the deviations appear random.

The available data for the enthalpy of sublimation are compared with the present model in Fig. 8. The primary data of Giauque and Egan²⁴ and Eucken and Donath,³¹ and the secondary data of Maass and Barnes¹⁹ are all represented within the experimental uncertainties. For the enthalpy of fusion, the equations of state of the solid and liquid phases yield a value of 8870 J mol⁻¹, while the three reported experimental values are (8050, 8340, and 7950) J mol⁻¹.^{19,20,40} The mean discrepancy of about 760 J mol⁻¹ cannot easily be ascribed to errors in the

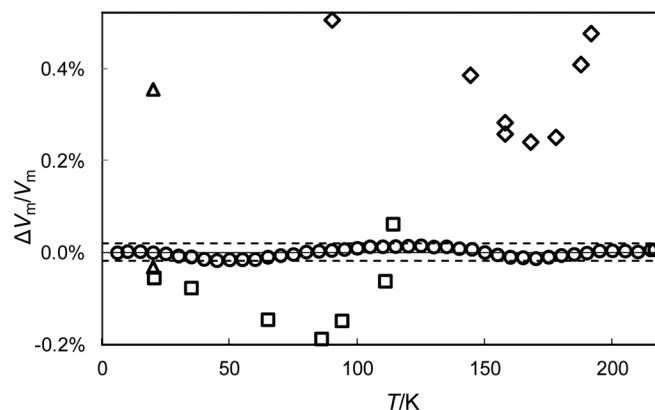


FIG. 5. Fractional deviations $\Delta V_{\text{m}}/V_{\text{m}}$ of the molar volume of solid CO₂ from the new equation of state on the sublimation curve as a function of temperature T : ◇, Maass and Barnes¹⁹ △, Heuse²¹ □, Keesom and Kohler,^{22,23,27} ○, Krupskii *et al.*²⁹ ---, uncertainty band for the equation of state.

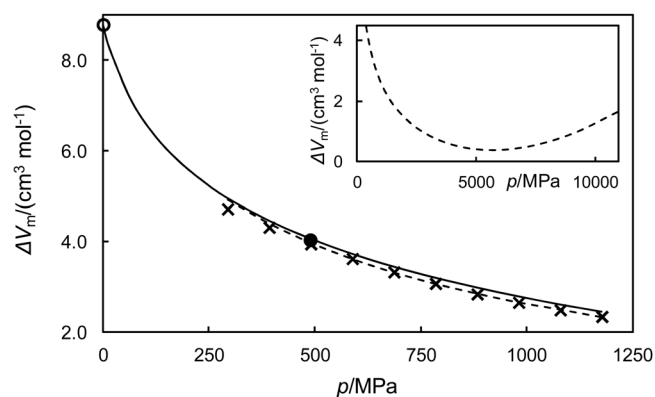


FIG. 6. Change in molar volume ΔV_m on melting: $\circ$, Krupskii *et al.*²⁹ with Ref. 5 for the liquid; $\bullet$, Giordano *et al.*¹⁷ with Ref. 5 for the liquid; $\times$, Bridgman;⁴⁷ —, solid equation of state with Ref. 5 for the liquid; ---, solid equation of state with Ref. 8 for the liquid. Insert: predicted behavior at high pressures.

new equation of state for the solid. Indeed, there is strong evidence to support the predicted enthalpy of the solid: the close agreement with experiment for $\Delta_s^g H_m$ at the normal sublimation point, and for the molar volume and the vapor pressures between the sublimation- and triple-point temperatures all point to errors below 100 J mol^{-1} for the enthalpy of the solid at the triple point. It also seems unlikely that the enthalpy of the gas phase at the triple point could be in significant error. Thus, either the three experimental values of $\Delta_s^g H_m$ share a similar systematic error or the enthalpy of vaporization at the triple point, which is 15.4 kJ mol^{-1} according to the fluid-phase equation of state, is in error by about 5%.

Figures 9 and 10 compare the primary and secondary sublimation-pressure data with the predictions of the new equation of state. All primary data are fitted to within a remarkably small band of $\pm 350 \text{ Pa}$. There is evidence of systematic differences between the results of different authors which, although small, exceed the estimated uncertainties, resulting in χ^2 values significantly above unity for these data sets. In the light of these differences, the small deviations compared with the model seen in Fig. 9 are entirely reasona-

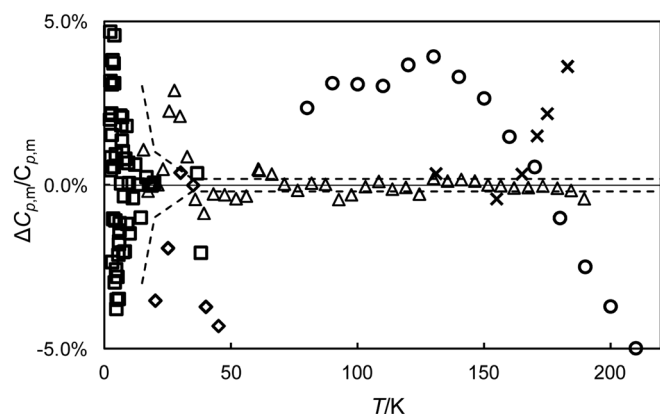


FIG. 7. Fractional deviations $\Delta C_{p,m}/C_{p,m}$ of isobaric molar heat capacity of solid CO_2 as a function of temperature T along the sublimation curve: $\square$, Manzhelii *et al.*;²⁶ $\triangle$, Giauque and Egan;²⁴ $\diamond$, Eucken;¹⁸ $\circ$, Eucken and Heuck;²⁰ $\times$, Maass and Barnes;¹⁹ ---, uncertainty band from Ref. 24.

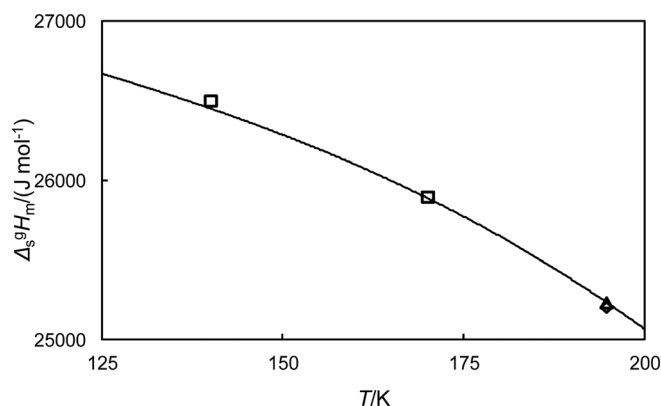


FIG. 8. Molar enthalpy of sublimation $\Delta_s^g H_m$ of CO_2 as a function of temperature T : $\triangle$, Giauque and Egan;²⁴ $\square$, Eucken and Donath;³¹ $\diamond$, Maass and Barnes;¹⁹ —, equation of state.

ble. The secondary data of Meyers and Van Dusen³² and Baughman *et al.*³⁵ also show rather small deviations, with Δ_{AAD} of 0.05% and 0.12%, respectively, while those of Giauque and Egan²⁴ deviate slightly more with $\Delta_{\text{AAD}} = 0.36\%$ and $\Delta_{\text{Bias}} = -0.30\%$. Figure 10 shows that the model is in fair agreement with the secondary data at temperatures below 150 K. The vapor pressures of Tickner and Lossing³³ are systematically lower than the predictions of the model by approximately twice the claimed experimental uncertainty with average deviations corresponding to a temperature error of approximately 0.6 K. Below 100 K, the data of Bryson *et al.*³⁶ scatter about the predicted curve as the sublimation pressure falls to $1.3 \times 10^{-8} \text{ Pa}$ at 70 K.

Figure 11 shows deviations of the primary and secondary melting-pressure data at pressures below 1.2 GPa. The primary data of Clusius *et al.*⁴⁹ are fitted to $\pm 1.2\%$, which is within the experimental uncertainty, while the secondary data^{47,48} are predicted to within $\pm 2\%$. Taken together, the experimental data suggest that the model systematically overestimates the melting pressure by approximately 1% at pressures between about 5 and 1200 MPa. However, it is not

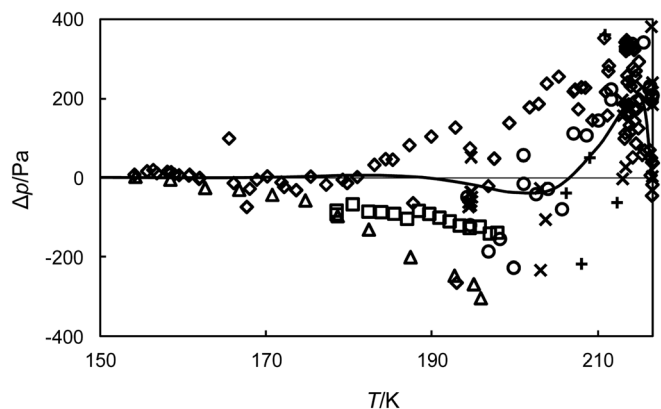


FIG. 9. Deviations Δp of experimental sublimation pressures of CO_2 at $T > 150 \text{ K}$ from the new equation of state for solid CO_2 with Ref. 5 for the vapor. $\circ$, Fernandez-Fassnacht and del Rio;³⁸ $\square$, Ambrose;³⁴ $\diamond$, Bilkadi *et al.*;³⁷ $\triangle$, Giauque and Egan;²⁴ $\times$, Meyers and Van Dusen;³² $+$, Baughman *et al.*;³⁵ —, Eq. (48).

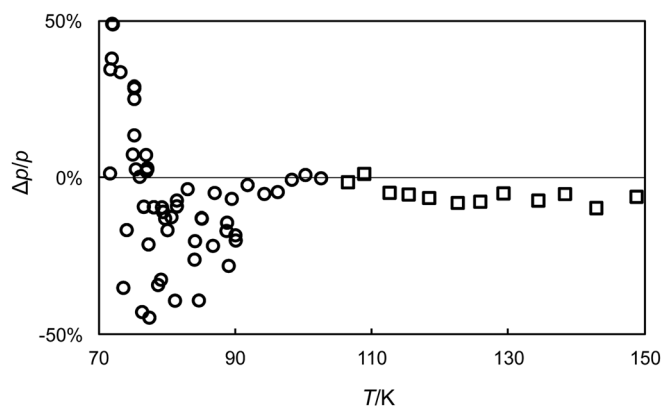


FIG. 10. Fractional deviations $\Delta p/p$ of experimental sublimation pressures of CO₂ at $T \leq 150$ K from the new equation of state for solid CO₂ with Ref. 5 for the vapor: $\square$, Tickner and Lossing;³³ $\circ$, Bryson *et al.*;³⁶ —, equation of state.

possible to attribute the deviations specifically to the new equation of state for the solid phase, as the calculated melting pressures are equally strongly dependent on the fluid-state equation of state.

Figures 12 to 14 compare experimental values of derivative properties with the predictions of the model. The thermal expansivity along the sublimation curve is shown in Fig. 12, and the agreement is mostly within the experimental uncertainties except for the results of Manzhelli *et al.*,²⁶ the only primary data considered for derivative properties, at very low temperatures and between 120 and 139 K. The secondary data of Krupskii *et al.*²⁹ are represented extremely well, even at very low temperatures. The isentropic compressibility along the sublimation curve is shown in Fig. 13. The disagreement between the different literature sources is clear here; the model tends to favor Manzhelli *et al.*²⁶ at higher temperatures and Yamashita and Kato³⁰ at lower temperatures. Finally, in Fig. 14, the isentropic bulk modulus measurements of Shimizu *et al.*⁴⁵ at $T = 300$ K are compared with the model. The experimental isotherm is essentially a linear function of pressure, while the predictions of the model exhibit some curvature. For these data, $\Delta_{\text{AAD}} = 9\%$, which is about twice the estimated experimental uncertainty.

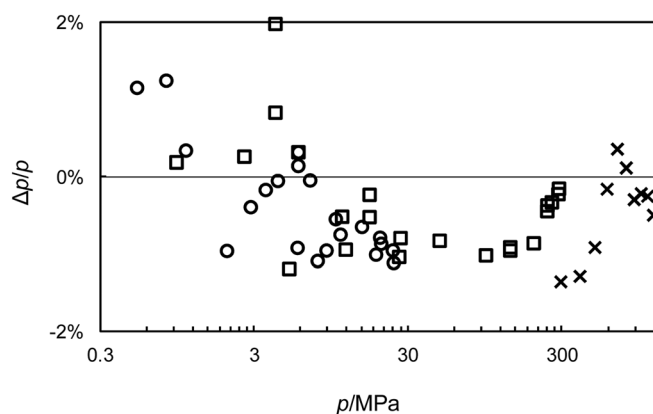


FIG. 11. Fractional deviations $\Delta p/p$ of experimental melting pressures of CO₂ from the new equation of state for solid CO₂ with Ref. 5 for the liquid: $\circ$, Clusius *et al.*;⁴⁹ $\square$, Michels *et al.*;⁴⁸ $\times$, Bridgman.⁴⁷

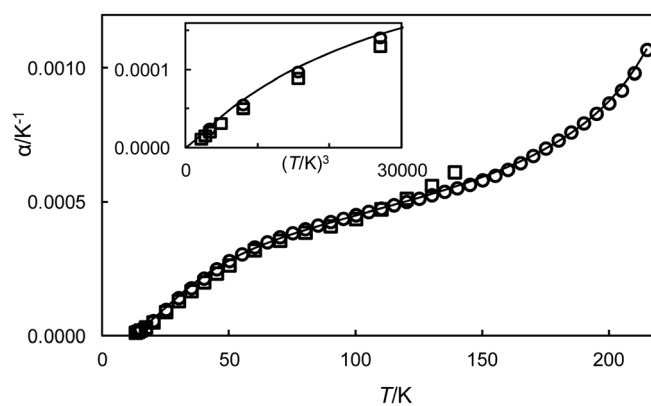


FIG. 12. Isobaric expansivity of solid CO₂ as a function of temperature T along the sublimation curve: $\circ$, Krupskii *et al.*;²⁹ $\square$, Manzhelli *et al.*;²⁶ —, equation of state. The insert shows α as a function of T^3 for $T \leq 31$ K.

Consideration was given to eliminating the most discrepant primary data sets. The isobaric heat-capacity data of Manzhelli *et al.*²⁶ has the largest χ^2 and it appears that the uncertainty stated by the authors does not encompass the scatter of their raw data; however, it may be reasonable for a smooth curve drawn through them. These data are in fact vital in determining the Debye temperature and so were retained. Their deviations appear random and the bias is zero. The three data sets for sublimation pressure have χ^2 values significantly above unity as a consequence of small systematic differences amongst them and the very small uncertainties assigned to these data. Eliminating any one data set might bias the fit and so retaining all three was felt to be best. The expansivity data of Manzhelli *et al.*²⁶ also have a quite large χ^2 , but eliminating these was found to have no significant influence on the final equation of state and so they too were retained.

Finally, in Fig. 15, the extrapolation behavior of the equation of state is considered in two further comparisons. The first is with the molar volumes reported by Liu⁴⁴ at $T = 296$ K and pressures between 11 and 51 GPa, which have a standard relative uncertainty of 0.6%. Liu did not observe a phase transition anywhere between 3 and 51 GPa, although it has been reported that the transition to phase III involves

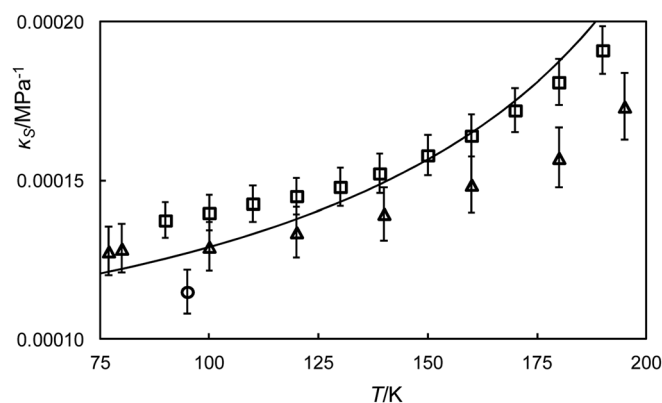


FIG. 13. Isentropic compressibility of solid CO₂ as a function of temperature T along the sublimation curve: $\square$, Manzhelli *et al.*;²⁶ $\triangle$, Yamashita and Kato;³⁰ $\circ$, Powell *et al.*;²⁸ —, equation of state.

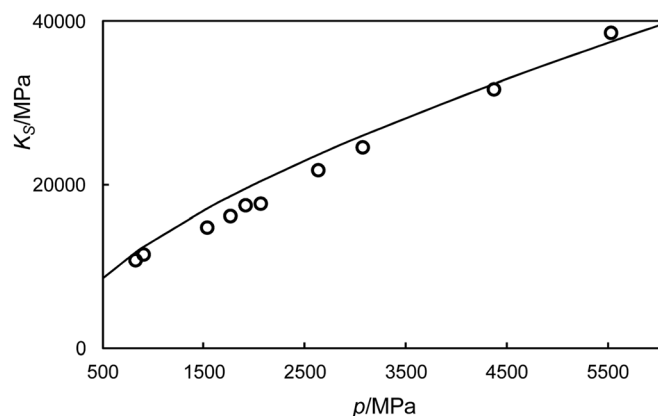


FIG. 14. Isentropic bulk modulus K_S of solid CO_2 as a function of pressure p at $T = 300$ K: $\circ$, Shimizu *et al.*;⁴⁵ —, equation of state.

negligible change in molar volume.⁶ Thus, these high-pressure data might relate to metastable states of either phase I or phase III. In any event, the experimental data agree with the present equation of state to within $\pm 1.2\%$ at pressures up to 25 GPa and only deviate by -3.6% at 51 GPa. The second comparison is with the Hugoniot data of Zubarev and Telegin,⁷⁴ which have a relative uncertainty of approximately 2% in $V_m(p)$. The results presented in the original paper require recalculation to correct for an error in the initial value of the density of the solid specimen at the normal sublimation point. With the correct value of 1.57 g cm^{-3} , the results shown in Fig. 15 are obtained. The computed curve is the isentrope passing through the initial state. The experimental data are in good agreement with the present model up to a pressure of about 20 GPa, after which increasing deviations are found, reaching 12% in the molar volume at $p = 68$ GPa. The calculated temperatures along the Hugoniot curve reach 960 K at the highest pressure attained experimentally.

8. Uncertainty of the Equation of State

Based on the comparison with experimental data presented in Figs. 4 and 5, the estimated relative uncertainty of the equa-

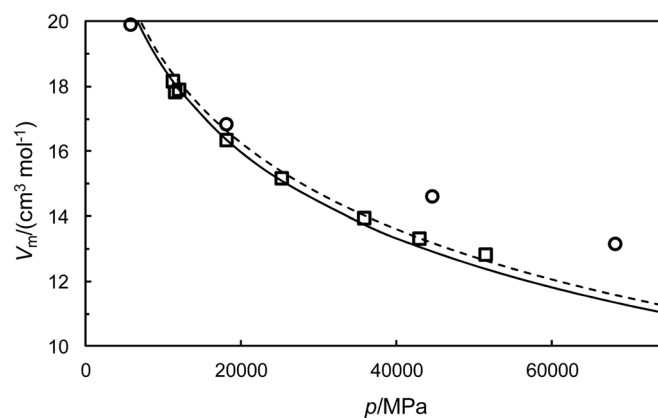


FIG. 15. Extrapolation behavior of the equation of state: $\square$, Liu ($T = 296$ K);⁴⁴ $\circ$, Hugoniot data of Zubarev and Telegin;⁷⁴ —, calculated for $T = 296$ K; ---, calculated Hugoniot curve.

tion of state for molar volume is 0.02% on the sublimation curve, increasing to 1.5% in the compressed solid. For isobaric heat capacity on the sublimation curve, the estimated relative uncertainties are: 5% for $T < 10$ K; 3% for $10 \leq T/K \leq 35$; and 0.5% for $T > 35$ K. It is not possible to provide a substantiated uncertainty estimate for the heat capacity of the compressed solid, as there are no experimental data with which to compare. However, in the limits of high temperature and high density, the lattice heat capacity is expected to approach the Dulong–Petit limit of $5R$ and the quasi-harmonic part of the equation of state conforms to this exactly. The anharmonic correction term in the equation of state lowers the lattice heat capacity only slightly, e.g., by about 0.5% at $T = 750$ K and $p = 10$ GPa. It is also noticeable that the contribution of the internal modes is very close to that found in the gas phase. Thus there is a reasonable expectation that the predicted constant-volume heat capacity should behave reasonably in the compressed solid at high temperatures.

9. Auxiliary Equations for the Melting and Sublimation Curves

9.1. Melting and sublimation pressure

The melting pressure of solid CO_2 cannot be reliably obtained by solving for equality of molar Gibbs energy between the new fundamental equation for the solid and the fluid equation of state of Span at Wagner⁵ at pressures above 800 MPa, which is the limit of validity of the latter. Ideally, a new and extended fluid-phase equation would be developed to address this issue but, in any event, it is useful to have an auxiliary equation from which the melting pressure may be obtained as a function of temperature. Span and Wagner used a simple quadratic polynomial in $(T/T_t - 1)$ to fit the melting curve at pressures up to 287 MPa. Abramson and Brown⁵² found that extending this to a cubic polynomial was sufficient to also fit the data at much higher pressures and the same conclusion is reached here. Accordingly, the equation for the melting pressure is

$$p = p_t \left[1 + d_1(T/T_t - 1) + d_2(T/T_t - 1)^2 + d_3(T/T_t - 1)^3 \right]. \quad (47)$$

A weighted fit to the primary and secondary melting-pressure data^{8,48,49,52} gave $d_1 = 2010.04$, $d_2 = 1704.40$, $d_3 = 171.695$, and $\Delta_{\text{AAD}} = 2.5\%$. The melting-curve data and their relative deviations from Eq. (47) are shown in Fig. 16, along with the predictions of the fundamental equations for the solid and fluid phases at pressures up to twice the nominal maximum pressures for the fluid-phase equation. The auxiliary equation is not as accurate as the fundamental equations at pressures below 300 MPa but, at higher pressures, it appears to do as well as could be expected given the scatter of the data and the systematic differences between the high-pressure data of Giordano *et al.*⁸ and Abramson and Brown.⁵² Also shown in Fig. 16 is the result of solving the new fundamental equation with the fluid-phase equation of state of Giordano *et al.*⁸ with

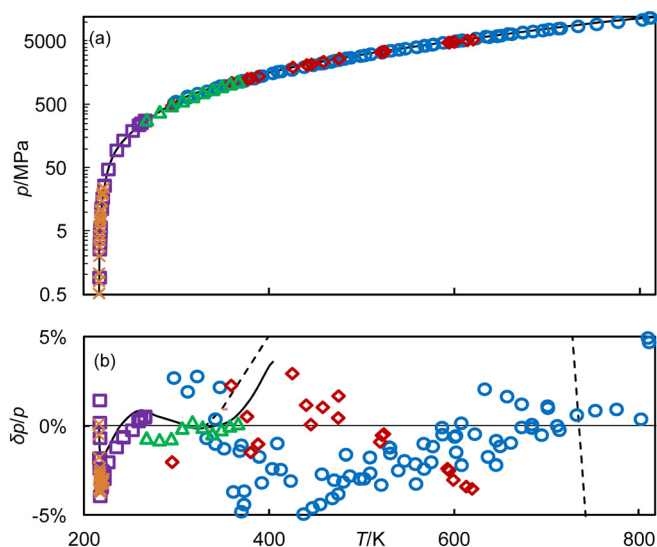


FIG. 16. (Color online) (a) Melting pressure p of solid CO₂ and (b) relative deviations $\delta p/p$ of melting pressures from the auxiliary equation: $\circ$, Giordano *et al.*;⁸ $\square$, Michels *et al.*;⁴⁸ $\times$, Clusius *et al.*;⁴⁹ $\diamond$, Abramson and Brown;⁵² $\triangle$, Bridgman; —, new equation of state for solid CO₂ solved with Ref. 5 for the liquid phase; - - -, new equation of state for solid CO₂ solved with Ref. 8 for the liquid phase.

G_m matched to that given by the Span–Wagner equation at $p=800$ MPa. Large deviations are found at temperatures above about 400 K. In part, the large errors reflect the rather small value of the change in molar volume upon melting found at high pressures (see Fig. 6).

In the case of the sublimation curve, the fundamental equations can be solved with confidence for equality of molar Gibbs energy to obtain the equilibrium pressure. Values computed in this way at temperatures between 150 and 216 K were fitted by an equation similar (but not identical) to that proposed by Span and Wagner,⁵

$$\ln(p/p_t) = (T_t/T) \left[d_4(1 - T/T_t) + d_5(1 - T/T_t)^{1.5} + d_6(1 - T/T_t)^{2.5} \right]. \quad (48)$$

The coefficients obtained were $d_4 = -14.97366$, $d_5 = 1.805904$, and $d_6 = -2.834695$. Equation (48) represents the sublimation pressures computed from the solid and fluid equations of state to within 0.04%, and it also provides a good representation of the experimental data, as can be seen in Fig. 9.

Equations (47) and (48) are both constrained by their functional forms to pass through the triple point.

9.2. Molar volume on the melting and sublimation curves

It is also convenient to have auxiliary equations for the molar volume along the melting and sublimation curves, e.g., to provide initial estimates for solving the equation of state on the phase boundaries. For this purpose, the molar volume was computed from the fundamental equation at a large number of

temperatures between 0 and 800 K at the pressures given by the auxiliary equations for the melting and sublimation pressures. The results were then fitted to separate closed-form equations above and below the triple point.

On the melting curve, the molar volume was correlated by the simple polynomial function,

$$V_m = V_{m,t} \left[1 + d_7(T/T_t - 1) + d_8(T/T_t - 1)^2 + d_9(T/T_t - 1)^3 \right], \quad (49)$$

where $V_{m,t} = 28.5589 \text{ cm}^3 \text{ mol}^{-1}$ is the molar volume of the solid at the triple-point temperature, $d_7 = 0.160433$, $d_8 = -0.018643$, and $d_9 = 0.001582$. This represents the values computed from the equation of state to within $\pm 0.07\%$.

In the case of the molar volume along the sublimation curve, the following equation represents V_m within $\pm 0.02\%$:

$$V_m = V_{00} \left[1 + \frac{d_{10} + d_{11}(T/T_0) + d_{12}(T/T_0)^2 + d_{13}(T/T_0)^3}{\exp(T_0/T) - 1} \right]. \quad (50)$$

Here, V_{00} is the molar volume at $T \rightarrow 0$ K (given in Table 6), $T_0 = 68.347$ K, $d_{10} = 0.00306956$, $d_{11} = 0.0243269$, $d_{12} = -0.00901163$, and $d_{13} = 0.00157151$.

Equation (49) is constrained by its functional form to agree with the equation of state at the triple point. Although Eq. (50) is not so constructed, the same constraint was imposed numerically when the parameters were optimized.

10. Discussion

It is interesting to examine the thermodynamic surface represented by the equation of state. Figure 17 shows the thermal properties: pressure, isothermal compressibility, isobaric expansivity, and thermal pressure coefficient, $\beta = (\partial p / \partial T)_V$; all computed along isotherms and on the

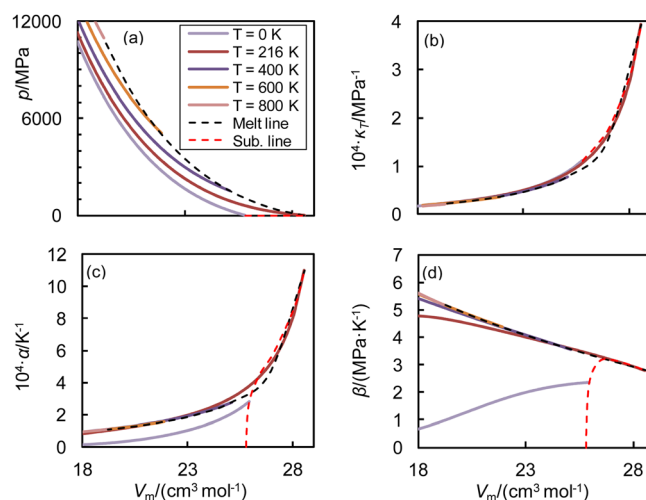


FIG. 17. (Color online) Thermal properties of solid phase I computed from the equation of state: (a) p - V_m isotherms; (b) isothermal compressibility κ_T ; (c) isobaric expansivity α ; (d) thermal pressure coefficient β .

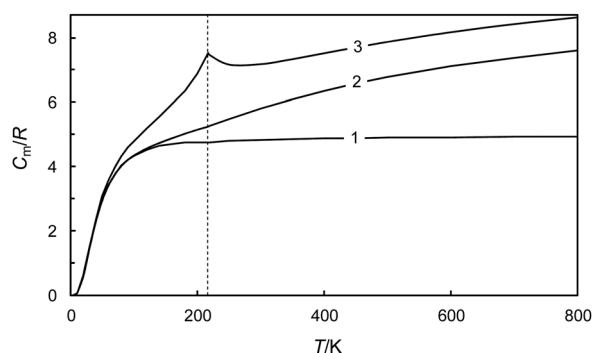


FIG. 18. Molar heat capacities C_m as functions of temperature along the sublimation curve ($T \leq T_t$) and melting curve ($T > T_t$). Curves: 1, lattice heat capacity $(C_{V,m} - C_{int})/R$; 2, isochoric molar heat capacity $C_{V,m}/R$; 3, isobaric molar heat capacity $C_{p,m}/R$. The dotted line marks the triple-point temperature.

phase boundaries over the full domain in which the equation of state is valid. The pressure isotherms appear approximately parallel, especially at high pressures. The isothermal compressibility is only very weakly dependent upon temperature but varies with molar volume, increasing rapidly in the approach to the triple point where the solid is in its most expanded state. The isobaric expansivity increases rapidly with temperature from zero at $T = 0$ K but at higher temperatures also becomes nearly athermal. Similar behavior is observed for $\beta = \alpha/\kappa_T$ which converges onto a single function of V_m with increasing T , indicating that the high-temperature (p, V_m, T) behavior is fundamentally rather simple.

Figure 18 shows the molar heat capacities of the solid, including the lattice heat capacity, $C_{V,m}$ and $C_{p,m}$, plotted as functions of temperature along the sublimation and melting curves. The lattice heat capacity, equal to $(C_{V,m} - C_{int})$, increases to reach $4.92R$ at $T = 800$ K, just below the Dulong–Petit limit of $5R$. The contribution of the internal modes takes $C_{V,m}$ to $7.60R$ at 800 K, while $C_{p,m}$ is $8.63R$ at the same temperature. There is a local maximum in $C_{p,m}$ at the triple point caused by the large values of the thermal expansivity in that most expanded state. A sharp peak is seen in Fig. 18 at the triple point because of the discontinuity there in the p - T locus along which $C_{p,m}$ is plotted.

The three Grüneisen parameters associated with the external lattice modes vary slowly with molar volume and span a range of approximately 1 to 4, whereas spectroscopic studies of these modes yield values close to 2.¹⁷ The thermal Grüneisen parameter, obtained from Eq. (39), is plotted in Fig. 19. The isotherms for this property do not converge at high temperatures because of the contribution of the internal modes to $C_{V,m}$. If the lattice heat capacity were used in place of the thermodynamic $C_{V,m}$, then the computed Grüneisen parameter would be approximately 2 at high temperatures.

The equation of state can also be used to explore metastable crystalline states. At $p = 0.1$ MPa, the solid is thermodynamically stable with respect to the vapor at temperatures up to about 195 K. However, the condition for mechanical stability, namely $(\partial^2 G_m / \partial V_m^2)_T < 0$, is satisfied for the meta-

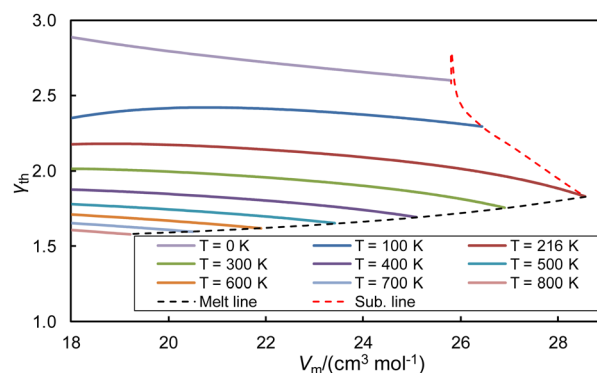


FIG. 19. (Color online) Thermal Grüneisen parameter γ_{th} computed from the equation of state.

stable solid to considerably higher temperatures and, according to the new equation of state, the absolute limit of mechanical stability is not reached until $T = 235$ K. As already mentioned, the equation of state may also be extrapolated to very high pressures to obtain properties of solid phase I in what are probably metastable states at $p > 12$ GPa.

The choice of the quasi-harmonic approximation leads to an equation of state in the form of a Helmholtz function with T and V_m as the independent variables. This approach was favored because of its theoretical basis, even though a Gibbs function such as that adopted for ice Ih by Feistel and Wagner,⁷⁵ with T and p as the independent variables, would be somewhat more convenient in application. In the present case, when T and p are specified, it is first necessary to solve the equation of state numerically for $V_m(T, p)$, but this is very easily achieved with a standard root finder (e.g., the Newton–Raphson method) and thereafter all properties follow from simple analytical expressions.

The cold curve used in this work does not correspond exactly to any of the simple intermolecular potential models that might be applied to the system. However, it can be represented approximately by a central Mie potential, constrained to give the equilibrium molar volume at $T = 0$ K, having an attractive exponent of 6 and a repulsive exponent of 10.87. However, this model does not match the accuracy of the present equation of state. The Vinet equation can also provide a reasonable representation of the cold curve.⁷⁰

11. Conclusions

The present work provides for the first time an equation of state for solid phase I of carbon dioxide from which all thermodynamic properties may be obtained with some confidence at pressures up to approximately 12 GPa and temperatures up to 800 K. The model is simple and easy to implement.

The equation of state is probably as good as it could be, given the data available at the present time. New measurements of the compression behavior at low temperatures would be useful, as there are no reliable data at $T < 295$ K, and for the compressibility along the sublimation curve. The latter would be best addressed by means of sound-speed measurements on single crystals. The equation of

state of the fluid is also a limitation in calculations of melting properties at $p > 800$ MPa, and it would be useful to develop an improved and extended fundamental equation valid up to the pressure of the (fluid + solid I + solid IV or VII) triple point.

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